

Introduction to Artificial Intelligence

Artificial Intelligence (AI) is a branch of computer science that focuses on creating systems capable of performing tasks that normally require human intelligence. These tasks include learning, reasoning, problem-solving, language understanding, decision-making, and recognizing patterns.

Machine Learning (ML) is a subset of Artificial Intelligence. Instead of being explicitly programmed for every task, machine learning algorithms learn from data and improve their performance over time. Common machine learning applications include spam email detection, recommendation systems, fraud detection, and image recognition.

Deep Learning is a specialized area of machine learning that uses artificial neural networks with multiple layers. Deep learning models have achieved outstanding performance in computer vision, speech recognition, natural language processing, and autonomous driving. These models require large datasets and significant computational resources for training.

Natural Language Processing (NLP) enables computers to understand, interpret, and generate human language. NLP is used in chatbots, virtual assistants, language translation systems, sentiment analysis, and question-answering systems. Modern NLP applications often rely on Large Language Models (LLMs), which are trained on vast amounts of text data.

Embeddings are numerical vector representations of text. They capture the semantic meaning of words, sentences, or documents, allowing similar pieces of text to be located using mathematical similarity. Vector databases store these embeddings and make semantic search efficient and scalable.

ChromaDB is an open-source vector database designed for storing and retrieving embeddings. It allows developers to create collections of documents, store metadata, and perform similarity searches. ChromaDB is widely used in Retrieval-Augmented Generation (RAG) systems, where relevant documents are retrieved before sending them to a language model.

Retrieval-Augmented Generation combines document retrieval with Large Language Models. Instead of relying only on the model's internal knowledge, RAG retrieves relevant information from external documents and includes it as context for the model. This approach improves factual accuracy and reduces hallucinations.

Ollama is a tool that enables developers to run Large Language Models locally on their own computers. It supports models such as Llama 3.2 and other open-source models. Running models locally provides better privacy, reduced latency, and offline capabilities.

Git is a version control system used by software developers to track changes in source code. It allows multiple developers to collaborate efficiently by maintaining a history of changes and supporting branching and merging workflows.

Python is one of the most popular programming languages for Artificial Intelligence and Machine Learning. It offers a rich ecosystem of libraries such as NumPy, Pandas, Scikit-learn, TensorFlow, and PyTorch. These libraries simplify data analysis, model development, and deployment.

Artificial Intelligence continues to transform industries including healthcare, education, finance, manufacturing, transportation, and entertainment. As AI technology advances, developers must also consider ethical issues such as fairness, transparency, privacy, and responsible use of data.